

1. (a)
 - i. Two coins are tossed, find the probability that a head and a tail are obtained.
 - ii. A die is rolled, find the probability that an prime number is obtained.
 - iii. A die is rolled and a coin is tossed, find the probability that the die shows an even number and the coin shows a tail.
 - (b) An ordinary die is thrown. Find the probability that the number obtained is
 - i. a multiple of 2.
 - ii. a factor of 6.
 - iii. less than 7.
 - (c) Two dice are rolled, find the probability that the sum of the two scores is
 - i. equal to 13
 - ii. equal to 4
 - iii. less than 10
 - iv. greater than 10
 - (d)
 - i. A card is drawn at random from a deck of cards. Find the probability of getting a diamond.
 - ii. A card is drawn at random from a deck of cards. Find the probability of getting the 3 of diamond.
 - (e) A jar contains 4 red marbles, 6 green marbles and 10 white marbles. If a marble is drawn from the jar at random, what is the probability that
 - i. this marble is green?
 - ii. this marble is not white?
2. (a) A die is rolled. What is the probability of getting
 - i. an odd number or even number?
 - ii. an even number or a number greater than 4.
- (b) In a class of 20 students, 4 of the 9 boys and 3 of the 11 girls are in the athletics team.
- A person from the class is chosen to participate in the "egg and spoon race" on Sports Day. Find the probability that the person chosen is
 - i. in the athletics team.
 - ii. female.
 - iii. a female member of the athletics team.
 - iv. a female or in the athletics team.
- (c) Events A and B are such that $P(A) = \frac{3}{10}$, $P(B) = \frac{2}{5}$ and $P(A \cap B) = \frac{1}{10}$. Find
 - i. $P(A \cap B^c)$.
 - ii. $P(A^c \cap B^c)$.

- (d) A card is drawn at random from an ordinary pack of playing cards. Find the probability that the card is
- a club or a diamond.
 - a club or a king.
3. (a) Events A and B are independent such that $P(A) = 0.3$, $P(B) = 0.5$. Find
- $P(A \cap B)$
 - $P(A \cup B)$
- (b) Events A and B are such that $P(A|B) = 0.4$, $P(B|A) = 0.25$ and $P(A \cap B) = 0.12$.
- Calculate the value of $P(B)$.
 - Explain why A and B are not independent.
 - Calculate the value of $P(A \cap B^c)$.
4. For the purposes of allocating first year students to different MAT 1110 lecture groups, a survey was randomly conducted on 600 students in the school of Humanities and Social Sciences of the University of Zambia. The results showed that:
- 60% took ECN 1115, 50% took BBA 1110, 45% took DEM 1110. In addition, 30% to ECN 1115 and BBA 1110, 28% took ECN 1115 and DEM 1110 and 25% took BBA 1110 and DEM 1110.
- 6% of the students took none of the these three courses.
- Find the probability that
- a student took none of the courses.
 - a student took all the courses.
 - a student took only one of the course.
5. (a) A die was thrown and the score was an odd number. What is the probability that it was a prime number?
- (b) In a certain University,

65% of the students are full-time students ,
 55% of the students are female ,
 35% of the students are male full-time students ,

Find the probability that

- a student chosen at random from all the students in the University is a part-time student.
 - a student chosen at random from all the students in the University is a female and a part-time student.
 - a student chosen at random from all the female students in the University is a part-time student.
- (c) Events X and Y are such that $P(X|Y) = \frac{2}{5}$, $P(Y) = \frac{1}{4}$ and $P(X) = \frac{1}{5}$. Find

- i. $P(Y|X)$
- ii. $P(Y \cap X)$
- iii. $P(Y \cup X)$

6. A diagnostic test has a probability of 0.95 of giving a positive result when applied to a person suffering from a certain disease, and a probability 0.10 of giving a (false) positive when applied a non-sufferer. It is estimated that 0.5% of the population are sufferers. Suppose that the test is now administered to a person about whom we have no relevant information relating to the disease (apart from the fact that he/she comes from this population). Calculate the following probability:

- (a) that the test result will be positive;
- (b) that, given a positive result, the person is a sufferer;
- (c) that, given a negative result, the person is a non-sufferer;
- (d) that the person will be misclassified.